

Cybersecurity Lab Report

Network Reconnaissance with Nmap

Target: Metasploitable2 (Docker) • Attacker: Kali Linux

Date: May 23, 2026

1. Introduction

This report documents a hands-on cybersecurity lab exercise focused on network reconnaissance using Nmap, one of the most widely used open-source tools for network discovery and security auditing. The goal of this first phase of the lab is to thoroughly enumerate a target machine — identifying active hosts, open ports, running services, software versions, and operating system fingerprints — without attempting any exploitation.

The lab environment consists of two virtual machines:

- Attacker machine: Kali Linux — a Debian-based distribution specifically designed for penetration testing and security research, equipped with a full suite of pre-installed security tools including Nmap and Zenmap.
- Target machine: Metasploitable2 — a deliberately vulnerable Linux system originally developed by Rapid7. In this lab it is deployed as a Docker container (tleemcjr/metasploitable2) running on a Linux Mint host, with several intentionally insecure services exposed on multiple ports.

The target container runs the following services: Apache2 (HTTP), OpenSSH, MySQL, ProFTPD, vsftpd, PostgreSQL, Tomcat 5.5, and Samba. These are started manually inside the container after launch, providing a realistic multi-service attack surface. Additional containers running alongside Metasploitable2 include DVWA, OWASP Juice Shop, Nagios, WordPress, nginx, and a Log4Shell-vulnerable application.

In this phase of the lab, only Nmap (and its graphical front-end Zenmap) is used. Future phases will introduce additional tools such as Metasploit, Burp Suite, and others to actively exploit the vulnerabilities discovered during this reconnaissance stage.

The techniques demonstrated in this report include:

- ARP scanning to discover live hosts on the local network
 - ICMP ping sweeps to confirm host availability
 - SYN stealth scans (-sS) to enumerate open TCP ports without completing the handshake
 - Service and version detection (-sV) to fingerprint running software
 - OS detection (-O) using TCP/IP stack fingerprinting
 - Nmap Scripting Engine (NSE) scripts (-A) for deeper enumeration of SMB, SSH, MySQL, and HTTP services
 - Saving scan results in multiple output formats (.nmap, .xml, .gnmap) and converting to HTML reports with xsltproc
 - Visual analysis using the Zenmap GUI (Ports/Hosts, Topology, Host Details tabs)
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2. Environment Setup

2.1 — Docker Infrastructure on Linux Mint Host

Before beginning the Kali reconnaissance, the Metasploitable2 Docker container and companion vulnerable application containers were confirmed to be running on the Linux Mint host. The following screenshot shows the Docker process list (`docker ps`) with all active containers and their port mappings.

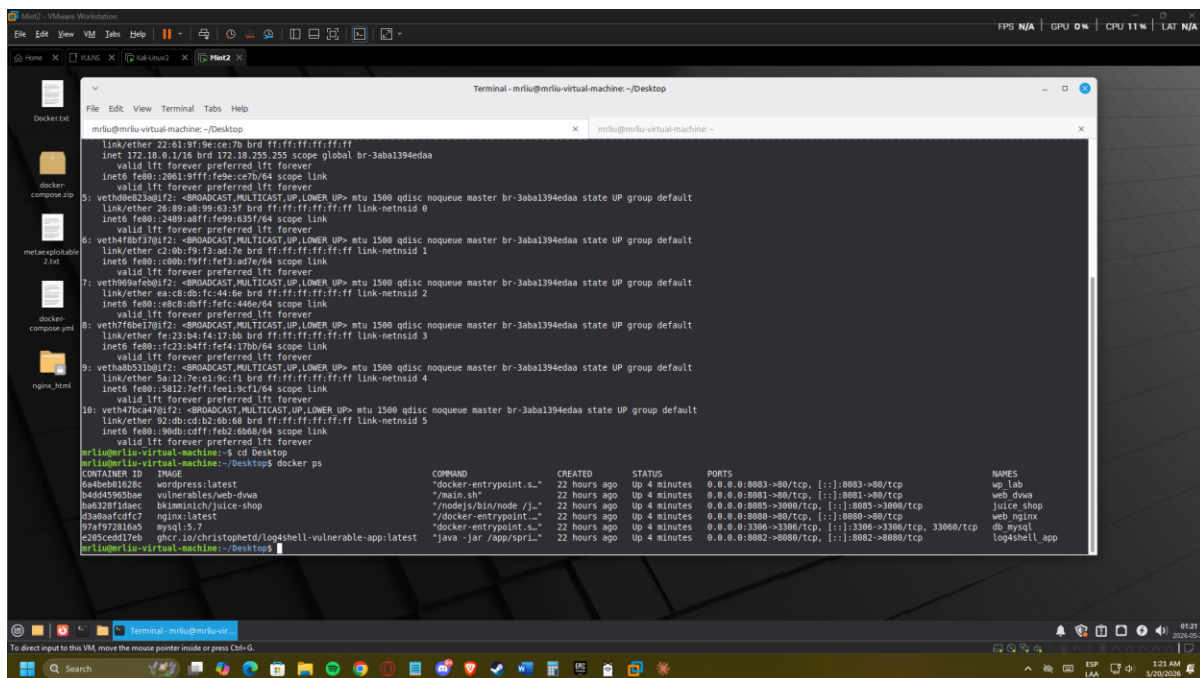


Figure 1 — Docker container list on Linux Mint host showing Metasploitable2, DVWA, Juice Shop, nginx, MySQL, WordPress, and Log4Shell containers with their port bindings.

The Metasploitable2 container was (re)launched using `docker rm -f metasploitable2` to remove any existing instance, followed by `docker run` with explicit port mappings. A `docker exec` session was then opened inside the container to manually start each vulnerable service.

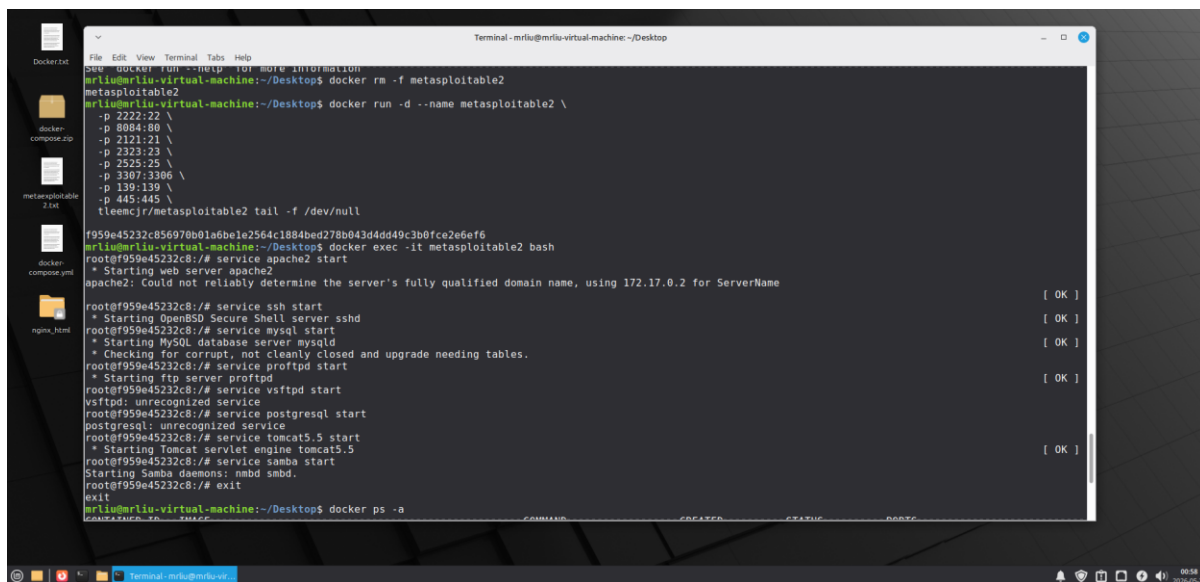


Figure 2 — Launching the Metasploitable2 container with port mappings (2222→22, 8084→80, 2121→21, etc.) and starting services: Apache2, SSH, MySQL, ProFTPD, vsftpd, PostgreSQL, Tomcat 5.5, and Samba.

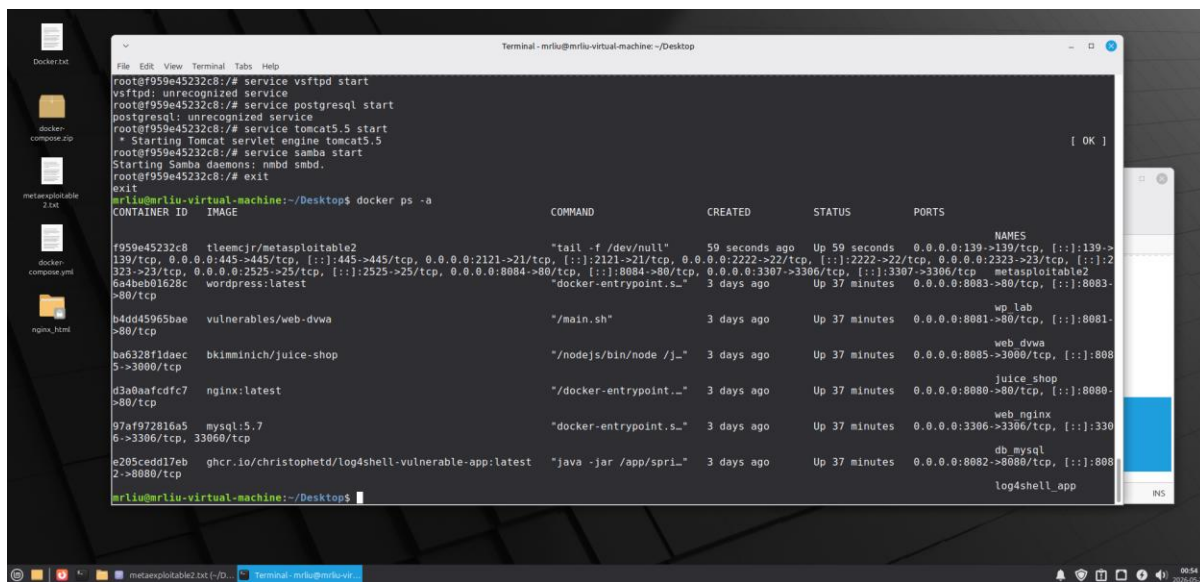


Figure 3 — Continuation of service startup output inside the Metasploitable2 container, confirming Tomcat 5.5 and Samba daemon (nmbd/smbd) started successfully.

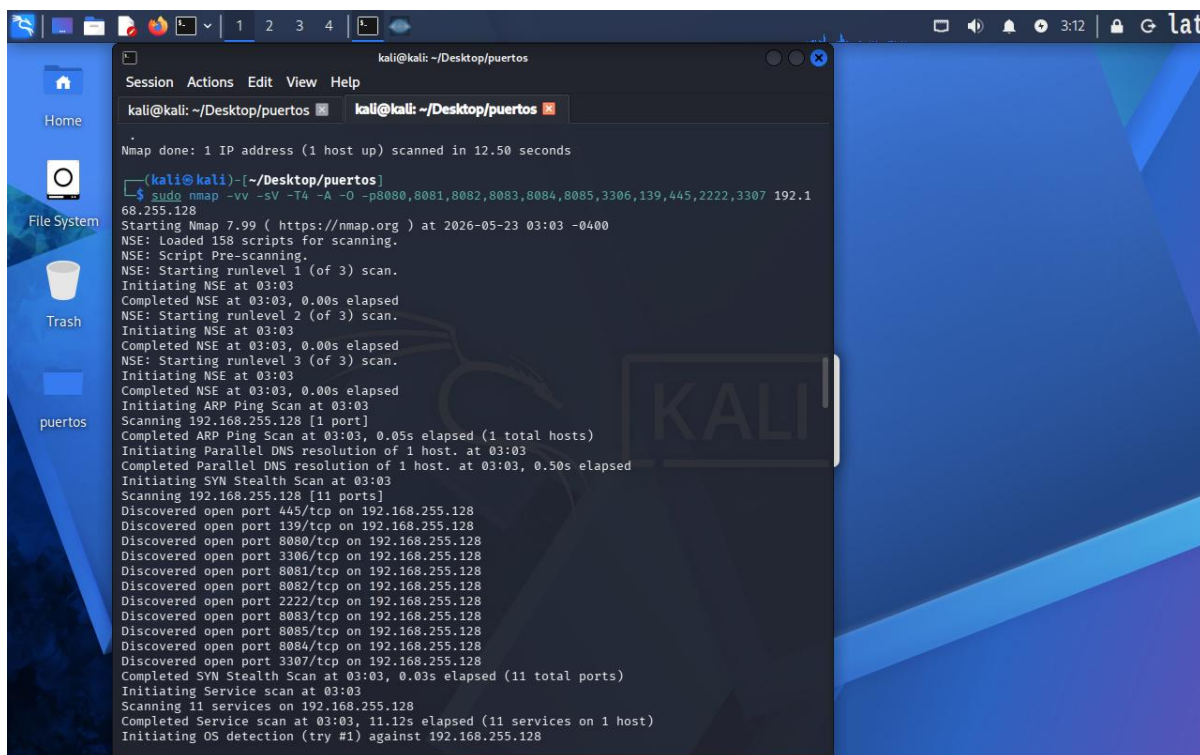
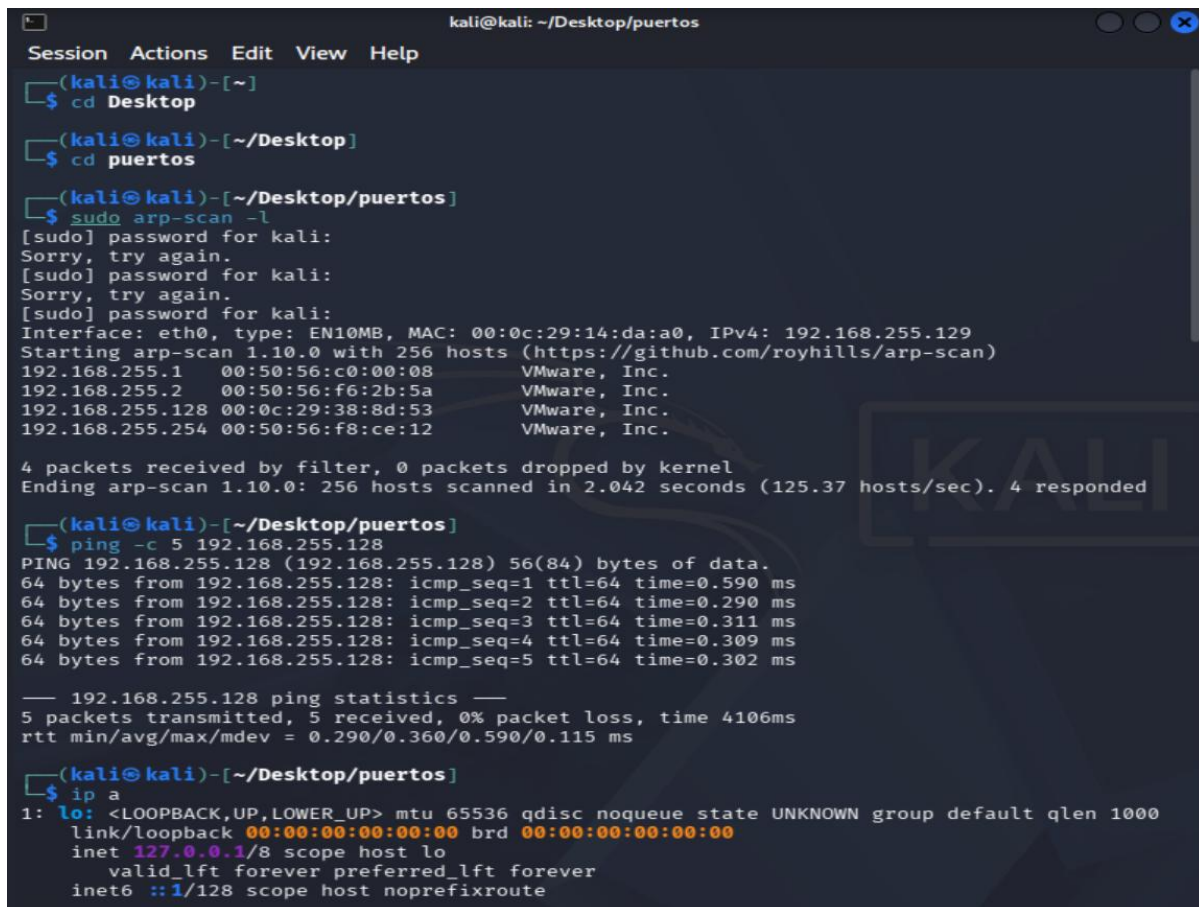


Figure 4 — Final `docker ps -a` output after starting all services, confirming the Metasploitable2 container is up with its complete port mapping table.

3. Network Reconnaissance with Nmap

3.1 — Host Discovery: ARP Scan and Ping

The reconnaissance phase began on the Kali Linux machine with an ARP scan (`sudo arp-scan -l`) to identify all live hosts on the local network segment. Four hosts were discovered, including the target at 192.168.255.128 (identified as a VMware adapter). Connectivity was then verified with a standard ICMP ping and `ip a` was run to confirm Kali's own network configuration.

A screenshot of a Kali Linux terminal window titled 'kali@kali: ~/Desktop/puertos'. The terminal shows the following commands and output:

```
(kali@kali)-[~]
$ cd Desktop
(kali@kali)-[~/Desktop]
$ cd puertos
(kali@kali)-[~/Desktop/puertos]
$ sudo arp-scan -l
[sudo] password for kali:
Sorry, try again.
[sudo] password for kali:
Sorry, try again.
[sudo] password for kali:
Interface: eth0, type: EN10MB, MAC: 00:0c:29:14:da:a0, IPv4: 192.168.255.129
Starting arp-scan 1.10.0 with 256 hosts (https://github.com/royhills/arp-scan)
192.168.255.1    00:50:56:c0:00:08    VMware, Inc.
192.168.255.2    00:50:56:f6:2b:5a    VMware, Inc.
192.168.255.128  00:0c:29:38:8d:53    VMware, Inc.
192.168.255.254  00:50:56:f8:ce:12    VMware, Inc.

4 packets received by filter, 0 packets dropped by kernel
Ending arp-scan 1.10.0: 256 hosts scanned in 2.042 seconds (125.37 hosts/sec). 4 responded

(kali@kali)-[~/Desktop/puertos]
$ ping -c 5 192.168.255.128
PING 192.168.255.128 (192.168.255.128) 56(84) bytes of data:
64 bytes from 192.168.255.128: icmp_seq=1 ttl=64 time=0.590 ms
64 bytes from 192.168.255.128: icmp_seq=2 ttl=64 time=0.290 ms
64 bytes from 192.168.255.128: icmp_seq=3 ttl=64 time=0.311 ms
64 bytes from 192.168.255.128: icmp_seq=4 ttl=64 time=0.309 ms
64 bytes from 192.168.255.128: icmp_seq=5 ttl=64 time=0.302 ms

— 192.168.255.128 ping statistics —
5 packets transmitted, 5 received, 0% packet loss, time 4106ms
rtt min/avg/max/mdev = 0.290/0.360/0.590/0.115 ms

(kali@kali)-[~/Desktop/puertos]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
```

Figure 5 — Kali terminal: `arp-scan` discovers 4 hosts on the 192.168.255.0/24 subnet; `ping` confirms 192.168.255.128 is reachable with 0% packet loss; `ip a` shows Kali's network interfaces.

3.2 — Initial Nmap Port Scan (Stealth SYN Scan)

An initial Nmap stealth SYN scan (`sudo nmap`) was run against 192.168.255.128 to discover which ports were open. Early scans without the correct port list showed filtered results because the Docker host was blocking unmapped ports. Once the scan was targeted at the exact ports forwarded by Docker, open services were revealed.

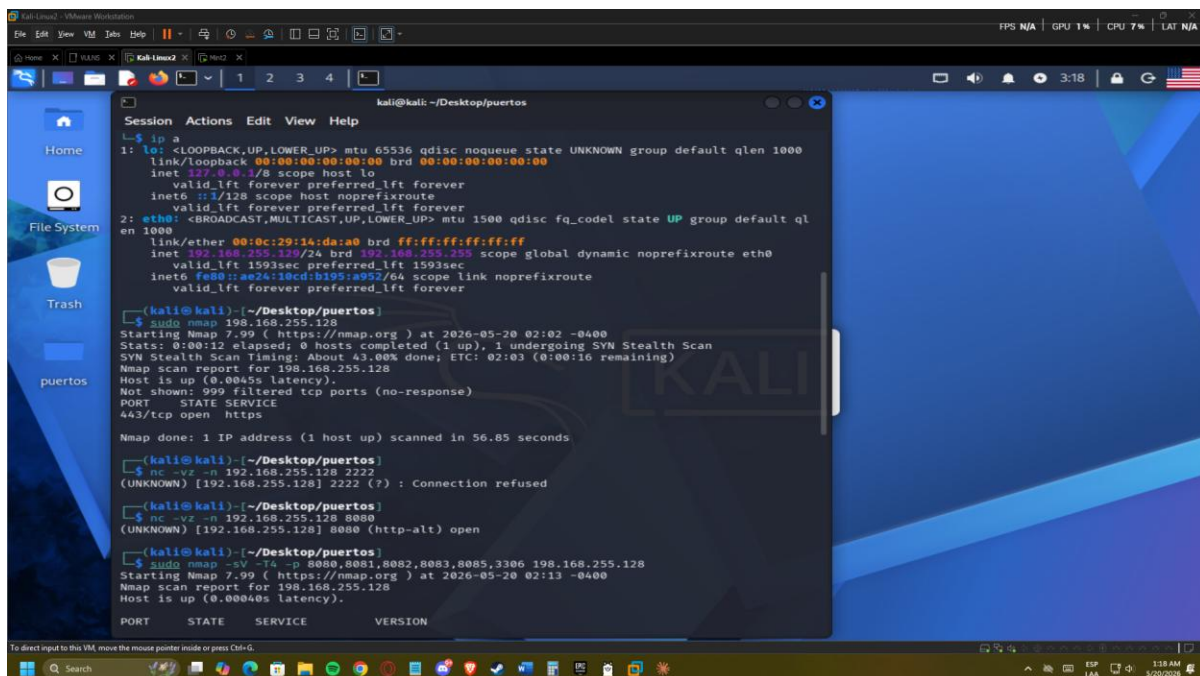


Figure 6 — Kali terminal showing early Nmap scan attempts against 198.168.255.128: ports 3306, 8080–8085 initially appear filtered. A subsequent scan on 192.168.255.128 reveals open ports including 3306/mysq and 8080–8085.

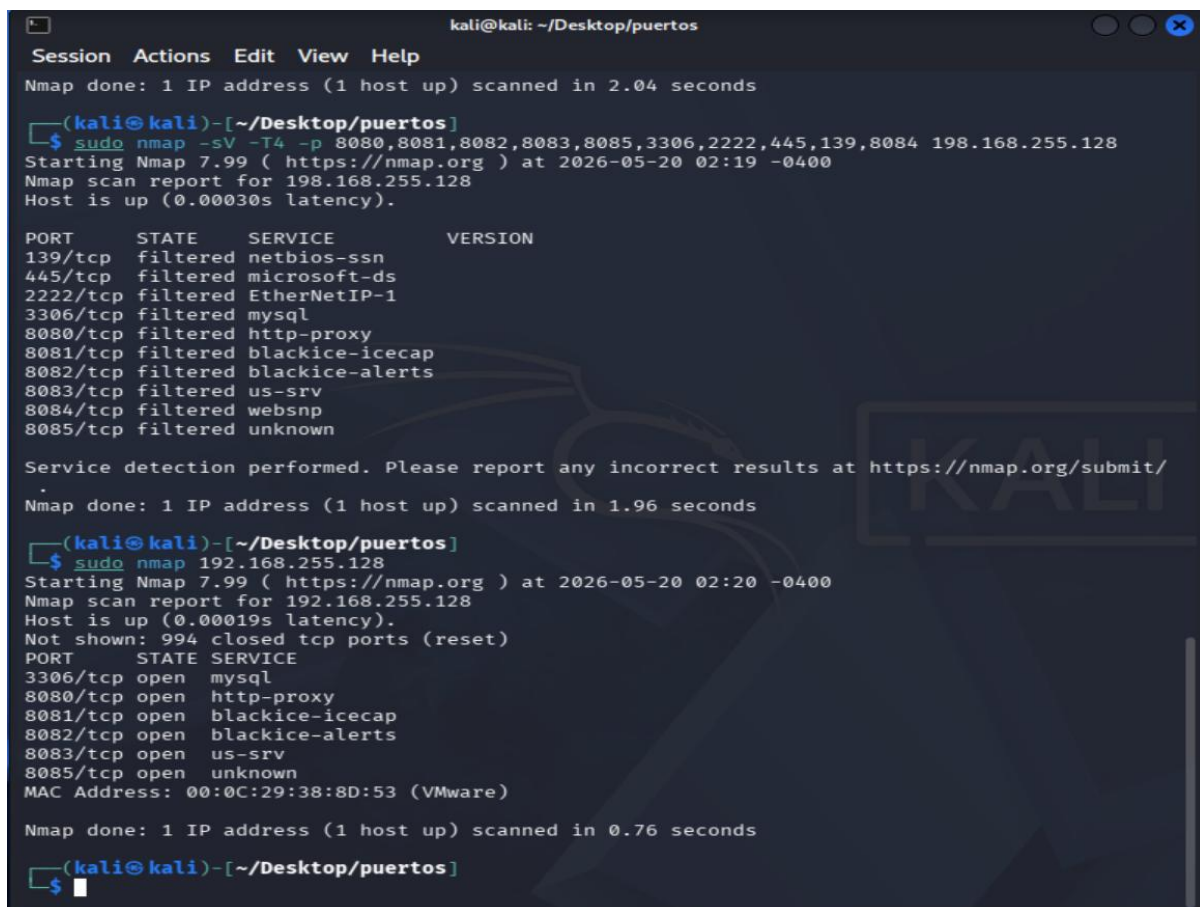


Figure 7 — Nmap scan results confirming all target ports are open on 192.168.255.128: 3306 (MySQL), 8080 (http-proxy), 8081 (blackice-icecap), 8082 (blackice-alerts), 2222, 8083, 8085, plus 139/445 (NetBIOS/SMB).

```

kali@kali: ~/Desktop/puertos
Session Actions Edit View Help
Nmap scan report for 198.168.255.128
Host is up (0.00040s latency).

PORT      STATE      SERVICE      VERSION
3306/tcp   filtered   mysql
8080/tcp   filtered   http-proxy
8081/tcp   filtered   blackice-icecap
8082/tcp   filtered   blackice-alerts
8083/tcp   filtered   us-srv
8085/tcp   filtered   unknown

Service detection performed. Please report any incorrect results at https://nmap.org/submit/
.
Nmap done: 1 IP address (1 host up) scanned in 2.04 seconds

(kali@kali)-[~/Desktop/puertos]
$ sudo nmap -sV -T4 -p 8080,8081,8082,8083,8085,3306,2222,445,139,8084 198.168.255.128
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-20 02:19 -0400
Nmap scan report for 198.168.255.128
Host is up (0.00030s latency).

PORT      STATE      SERVICE      VERSION
139/tcp    filtered   netbios-ssn
445/tcp    filtered   microsoft-ds
2222/tcp   filtered   EtherNetIP-1
3306/tcp   filtered   mysql
8080/tcp   filtered   http-proxy
8081/tcp   filtered   blackice-icecap
8082/tcp   filtered   blackice-alerts
8083/tcp   filtered   us-srv
8084/tcp   filtered   websnp
8085/tcp   filtered   unknown

Service detection performed. Please report any incorrect results at https://nmap.org/submit/
.
Nmap done: 1 IP address (1 host up) scanned in 1.96 seconds

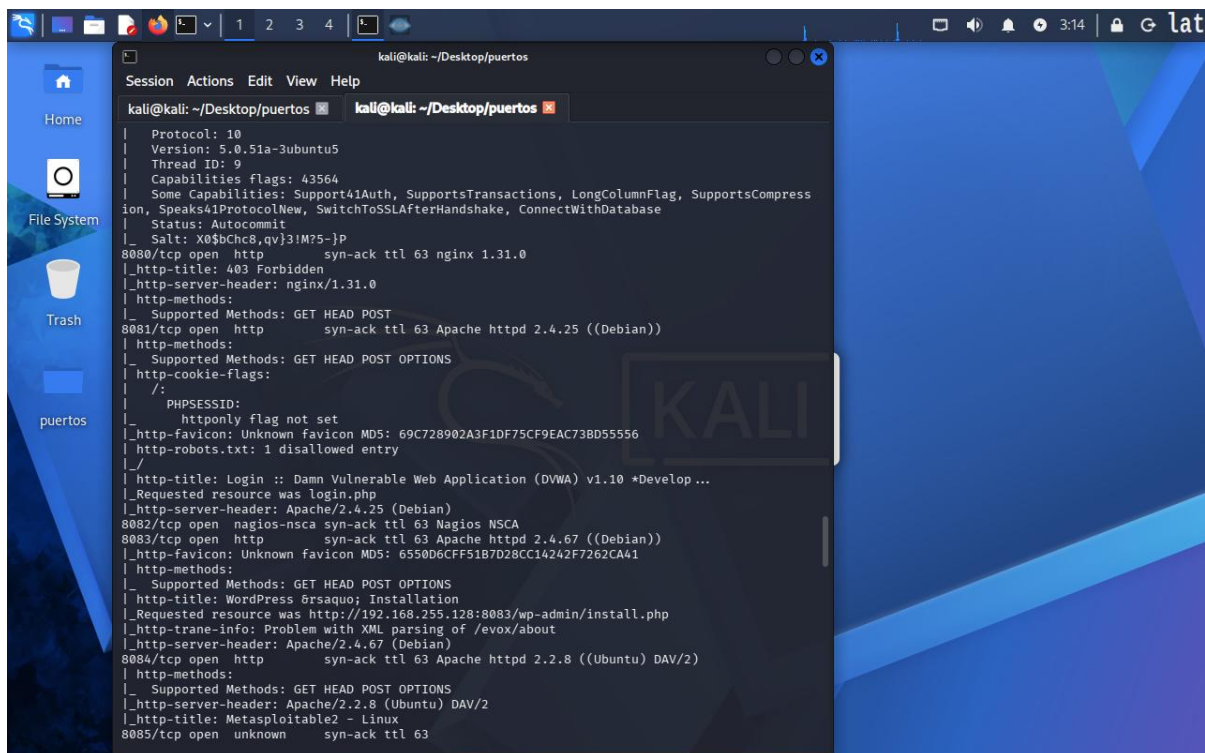
(kali@kali)-[~/Desktop/puertos]
$ sudo nmap 192.168.255.128
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-20 02:20 -0400
Nmap scan report for 192.168.255.128
Host is up (0.00019s latency).
Not shown: 994 closed tcp ports (reset)

```

Figure 8 — Kali terminal (Linux Mint VM view): verbose Nmap scan (`-vv -sV -T4 -A -D`) running against 192.168.255.128 with all target ports specified, showing real-time port discovery progress.

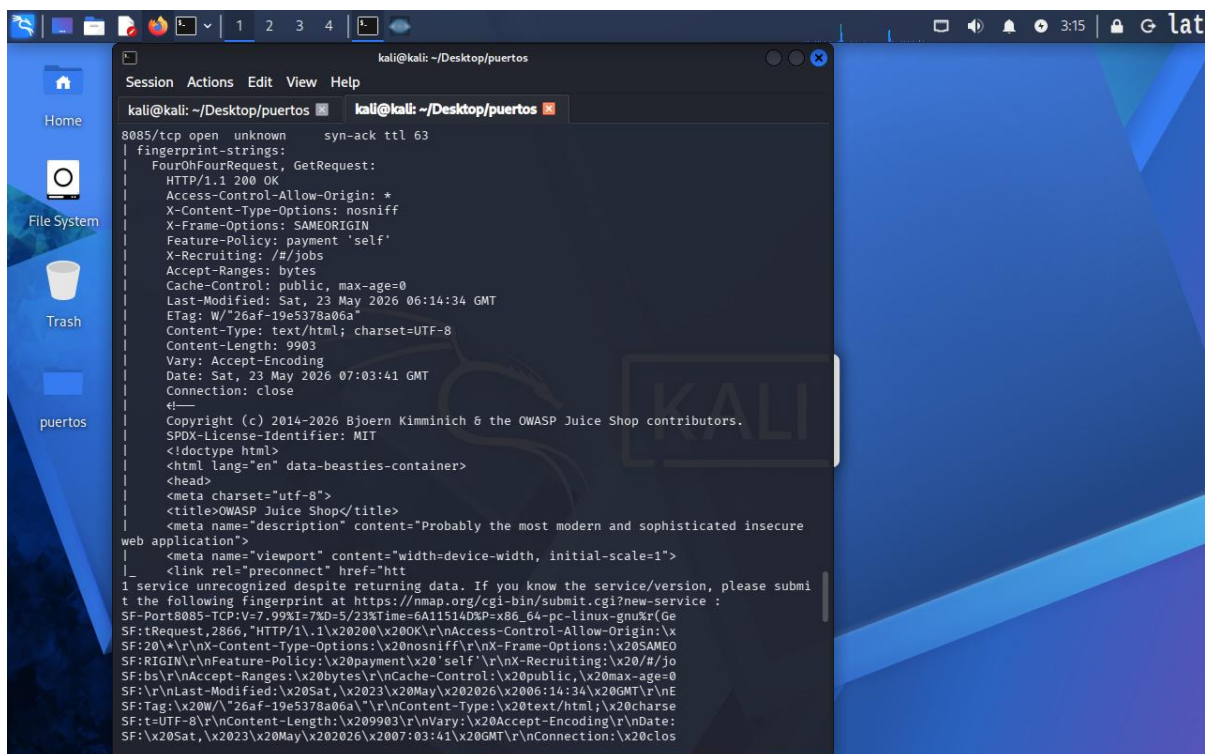
3.3 — Full Aggressive Scan with Service & OS Detection

A comprehensive Nmap scan was executed using aggressive detection flags: ``-vv -sV -T4 -A -O`` combined with the ``-D`` decoy option and all known Docker-mapped ports. The NSE scripts automatically enumerated SSH host keys, MySQL capabilities and SSL certificates, HTTP server headers and web application fingerprints, SMB security settings, and OS fingerprints.



```
kali@kali: ~/Desktop/puertos
Session Actions Edit View Help
kali@kali: ~/Desktop/puertos
| Protocol: 10
| Version: 5.0.51a-3ubuntu5
| Thread ID: 9
| Capabilities flags: 43564
| Some Capabilities: Support41Auth, SupportsTransactions, LongColumnFlag, SupportsCompress
| ion, Speaks41ProtocolNew, SwitchToSSLAfterHandshake, ConnectWithDatabase
| Status: Autocommit
| Salt: X0$bChc8,qv}3!M75-j}P
8080/tcp open  http      syn-ack ttl 63 nginx 1.31.0
|_ http-title: 403 Forbidden
|_ http-server-header: nginx/1.31.0
|_ http-methods:
|_ Supported Methods: GET HEAD POST
8081/tcp open  http      syn-ack ttl 63 Apache httpd 2.4.25 ((Debian))
|_ http-methods:
|_ Supported Methods: GET HEAD POST OPTIONS
|_ http-cookie-flags:
|_ /:
|_ PHPSESSID:
|_ httponly flag not set
|_ http-favicon: Unknown favicon MD5: 69C728902A3F1DF75CF9EAC73BD55556
|_ http-robots.txt: 1 disallowed entry
|_ /:
|_ http-title: Login :: Damn Vulnerable Web Application (DVWA) v1.10 *Develop...
|_ Requested resource was login.php
|_ http-server-header: Apache/2.4.25 (Debian)
8082/tcp open  nagios-nsc syn-ack ttl 63 Nagios NSCA
8083/tcp open  http      syn-ack ttl 63 Apache httpd 2.4.67 ((Debian))
|_ http-favicon: Unknown favicon MD5: 6550D6CFF51B7D28CC14242F7262CA41
|_ http-methods:
|_ Supported Methods: GET HEAD POST OPTIONS
|_ http-title: WordPress &rsquo; Installation
|_ Requested resource was http://192.168.255.128:8083/wp-admin/install.php
|_ http-trane-info: Problem with XML parsing of /evox/about
|_ http-server-header: Apache/2.4.67 (Debian)
8084/tcp open  http      syn-ack ttl 63 Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_ http-methods:
|_ Supported Methods: GET HEAD POST OPTIONS
|_ http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
|_ http-title: Metasploitable2 - Linux
8085/tcp open  unknown   syn-ack ttl 63
```

Figure 11 — HTTP service enumeration: port 8080 runs nginx 1.31.0 (403 Forbidden); port 8081 serves the DVWA (Damn Vulnerable Web Application) login page via Apache 2.4.25; port 8082 shows Nagios NSCA; port 8083 exposes a WordPress installation page; port 8084 serves Apache 2.2.8 (Ubuntu) with DAV/2 enabled.



```
kali@kali: ~/Desktop/puertos
Session Actions Edit View Help
kali@kali: ~/Desktop/puertos
8085/tcp open  unknown   syn-ack ttl 63
|_ fingerprint-strings:
|_ FourOhFourRequest, GetRequest:
|_ HTTP/1.1 200 OK
|_ Access-Control-Allow-Origin: *
|_ X-Content-Type-Options: nosniff
|_ X-Frame-Options: SAMEORIGIN
|_ Feature-Policy: payment 'self'
|_ X-Recruiting: /#/jobs
|_ Accept-Ranges: bytes
|_ Cache-Control: public, max-age=0
|_ Last-Modified: Sat, 23 May 2026 06:14:34 GMT
|_ ETag: W/"26af-19e5378a06a"
|_ Content-Type: text/html; charset=UTF-8
|_ Content-Length: 9903
|_ Vary: Accept-Encoding
|_ Date: Sat, 23 May 2026 07:03:41 GMT
|_ Connection: close
|_ <!--
|_ Copyright (c) 2014-2026 Bjorn Kimminich & the OWASP Juice Shop contributors.
|_ SPOX-License-Identifier: MIT
|_ <!doctype html>
|_ <html lang="en" data-beasties-container>
|_ <head>
|_ <meta charset="utf-8">
|_ <title>OWASP Juice Shop</title>
|_ <meta name="description" content="Probably the most modern and sophisticated insecure
web application">
|_ <meta name="viewport" content="width=device-width, initial-scale=1">
|_ <link rel="preconnect" href="htt
1 service unrecognized despite returning data. If you know the service/version, please submit
the following fingerprint at https://nmap.org/cgi-bin/submit.cgi?new-service :
SF:Port8085-TCP:V=7.99I=7&D=5/23Time=6A115140XP=x86_64-pc-linux-gnu&r(Ge
SF:Request,2866,"HTTP/1.1x20200x200K\r\nAccess-Control-Allow-Origin:\x
SF:20(\r\nX-Content-Type-Options:\x20nosniff\r\nX-Frame-Options:\x20SAMEO
SF:RIGIN\r\nFeature-Policy:\x20payment\x20self\r\nX-Recruiting:\x20/#/jo
SF:bs\r\nAccept-Ranges:\x20bytes\r\nCache-Control:\x20public,\x20max-age=0
SF:\r\nLast-Modified:\x20Sat,\x2023\x20May\x202026\x2006:14:34\x20GMT\r\nE
SF:Tag:\x20W/\x26af-19e5378a06a\r\nContent-Type:\x20text/html;\x20charse
SF:t:UTF-8\r\nContent-Length:\x209903\r\nVary:\x20Accept-Encoding\r\nDate:
SF:\x20Sat,\x2023\x20May\x202026\x2007:03:41\x20GMT\r\nConnection:\x20clos
```

Figure 12 — Port 8085 HTTP fingerprint revealing the OWASP Juice Shop application: HTTP headers include X-Frame-Options: SAMEORIGIN, Feature-Policy, and HTML source confirming the Juice Shop title and description.

```

kali@kali: ~/Desktop/puertos
Session Actions Edit View Help

kali@kali: ~/Desktop/puertos kali@kali: ~/Desktop/puertos
OS::TS=A)OPS(01=M5B4ST11NWA%02=M5B4ST11NWA%03=M5B4NNT11NWA%04=M5B4ST11NWA%0
OS::S=M5B4ST11NWA%06=M5B4ST11)WIN(W1=FE88%W2=FE88%W3=FE88%W4=FE88%W5=FE88%W6
OS::FE88)ECN(R=Y%DF=Y%T=3F%W=FAF%00=M5B4NNSNWA%CC=Y%Q=)T1(R=Y%DF=Y%T=3F%S=0
OS::%A=S+%F=AS%RD=0%Q=)T2(R=N)T3(R=N)T4(R=Y%DF=Y%T=3F%W=0%S=AS%Z%F=R%O%RD=
OS::0%Q=)T5(R=Y%DF=Y%T=40%W=0%S=Z%AS+S+%F=AR%O%RD=0%Q=)T6(R=Y%DF=Y%T=40%W=0
OS::S=AS%Z%F=R%O%RD=0%Q=)T7(R=Y%DF=Y%T=40%W=0%S=Z%AS+S+%F=AR%O%RD=0%Q=)U1(
OS::R=Y%DF=N%T=40%IPL=164%UN=0%IPL=G%RID=G%RIPCK=G%RUCK=G%RUD=G)J1E(R=Y%DFI=
OS::N%T=40%CD=S)

Uptime guess: 31.079 days (since Wed Apr 22 01:09:59 2026)
Network Distance: 1 hop
TCP Sequence Prediction: Difficulty=260 (Good luck!)
IP ID Sequence Generation: All zeros
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
|_ smb-os-discovery:
|   OS: Unix (Samba 3.0.20-Debian)
|   Computer name: f959e45232c8
|   NetBIOS computer name:
|   Domain name:
|   FQDN: f959e45232c8
|   System time: 2026-05-23T03:03:47-04:00
|_ smb-security-mode:
|   account used: <blank>
|   authentication level: user
|   challenge response: supported
|   message signing: disabled (dangerous, but default)
|_ clock-skew: mean: 1h59m59s, deviation: 2h49m42s, median: 0s
|_ smb2-security-mode: Couldn't establish a SMBv2 connection.
|_ p2p-conficker:
|   Checking for Conficker.C or higher ...
|   Check 1 (port 21451/tcp): CLEAN (Couldn't connect)
|   Check 2 (port 45408/tcp): CLEAN (Couldn't connect)
|   Check 3 (port 27425/udp): CLEAN (Timeout)
|   Check 4 (port 12003/udp): CLEAN (Failed to receive data)
|_ 0/4 checks are positive: Host is CLEAN or ports are blocked
|_ smb2-time: Protocol negotiation failed (SMB2)

TRACEROUTE
HOP RTT ADDRESS

```

Figure 13 — Continuation of Juice Shop HTTP source fingerprint (raw response body) and OS detection output: OS type detected as Linux 4.x|5.x, CPE: cpe:/o:linux:linux_kernel, TCP/IP fingerprint string.

3.4 — SMB Enumeration and Traceroute

NSE scripts interrogated the SMB service on ports 139/445 and reported Samba version, computer name, domain, system time, and security configuration. The p2p-conficker check returned all-clean results. The traceroute confirmed the target is exactly 1 hop away on the local LAN.

4. Zenmap GUI Analysis

The same Nmap command was also run through Zenmap, the official graphical front-end for Nmap. Zenmap provides tabbed views that make it easier to navigate and interpret scan results visually.

4.1 — Nmap Output Tab

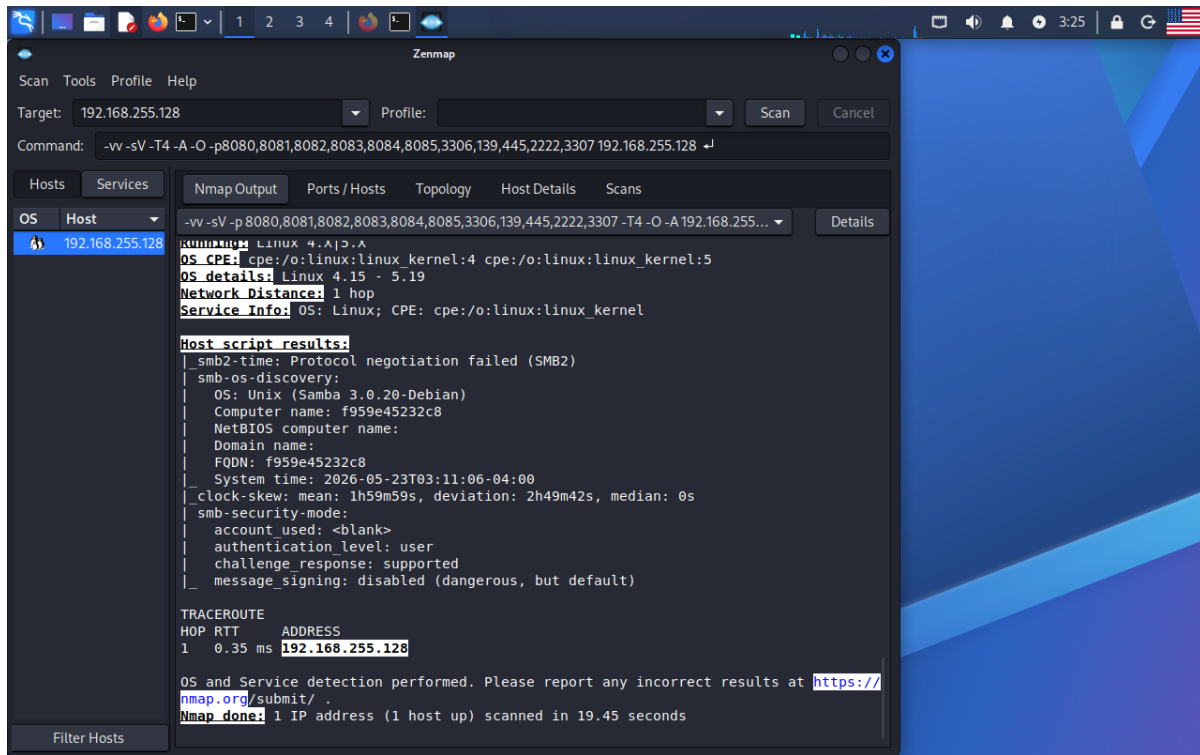


Figure 16 — Zenmap — Nmap Output tab: full scrollable text output of the aggressive scan, showing the same results as the CLI but within the Zenmap interface. Target: 192.168.255.128.

4.2 — Ports / Hosts Tab

The Ports/Hosts tab provides a clean tabular view of every discovered open port with its protocol, state, service name, and detected version string, making it easy to quickly assess the attack surface.

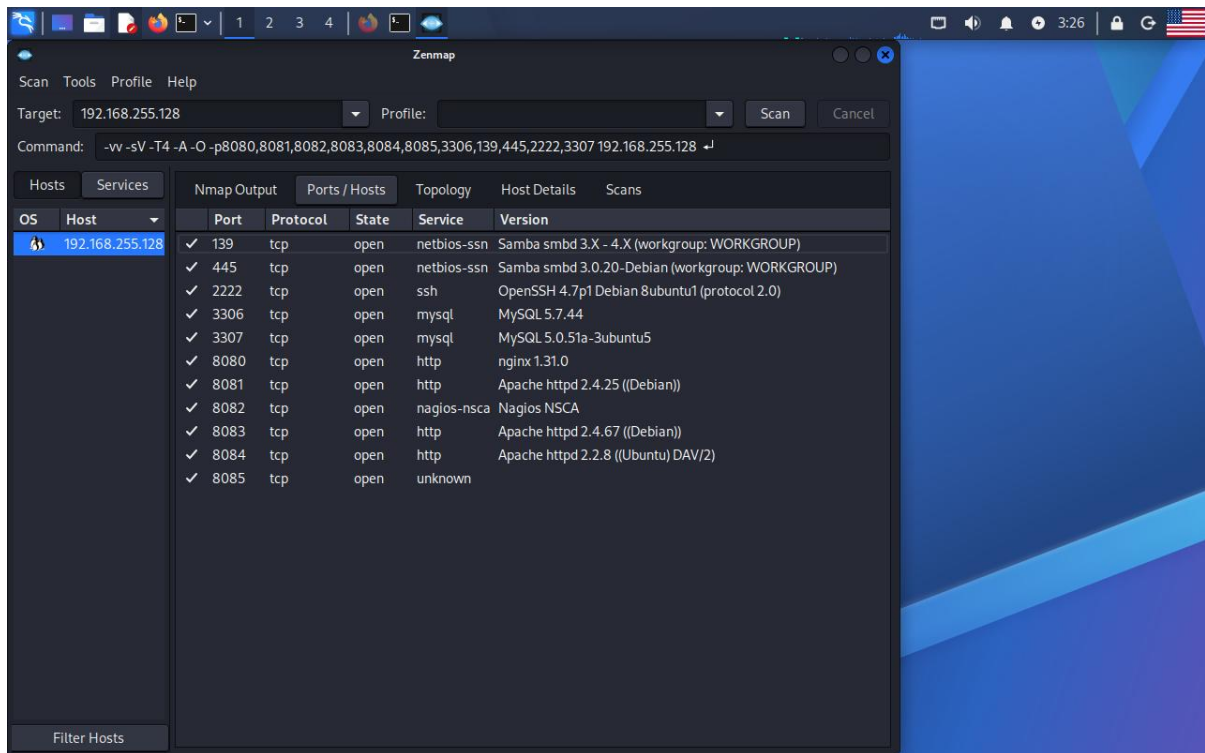


Figure 17 — Zenmap Ports/Hosts tab listing all 11 open TCP ports: 139 (netbios-ssn, Samba 3.x–4.x), 445 (netbios-ssn, Samba 3.0.20-Debian), 2222 (ssh, OpenSSH 4.7p1), 3306 (mysql, 5.7.44), 3307 (mysql, 5.0.51a), 8080 (http, nginx 1.31.0), 8081 (http, Apache 2.4.25 Debian), 8082 (nagios-nsc), 8083 (http, Apache 2.4.67), 8084 (http, Apache 2.2.8 Ubuntu DAV/2), 8085 (unknown).

4.3 — Topology Tab

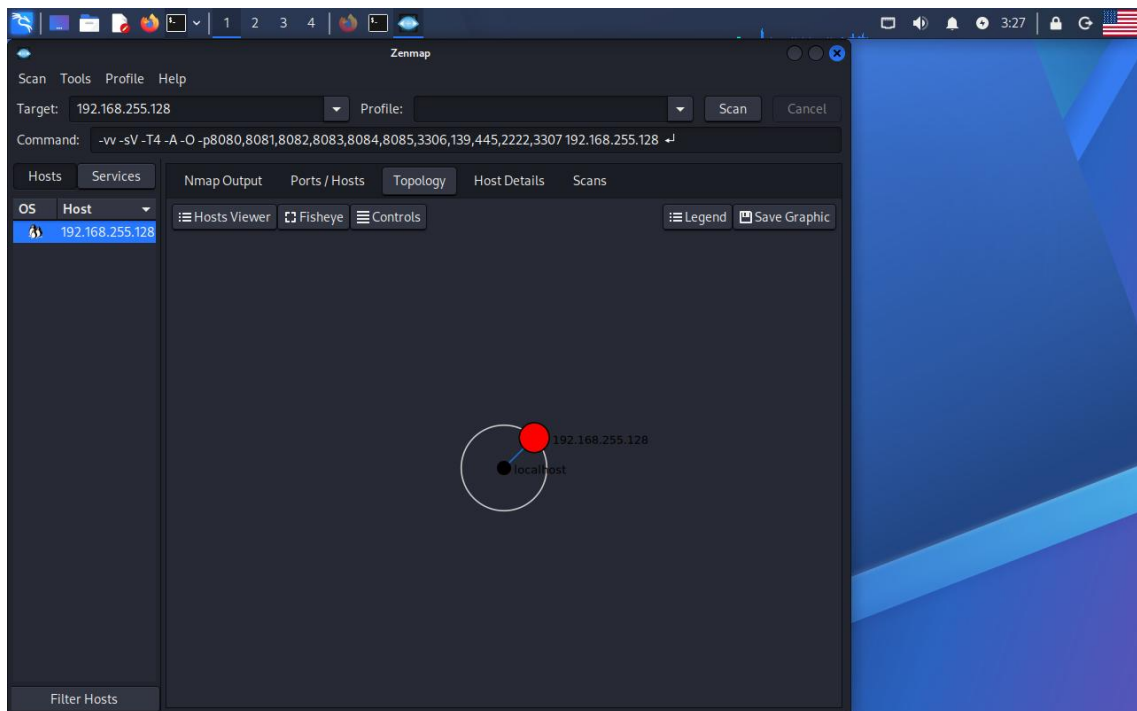


Figure 18 — Zenmap Topology tab: interactive network graph showing the relationship between localhost (Kali) and the target host 192.168.255.128. The red node represents the scanned target directly connected to the local machine.

4.4 — Host Details Tab

The Host Details tab aggregates all information Nmap collected about the target into a structured overview including host status, addresses, uptime, operating system, and network fingerprinting data.

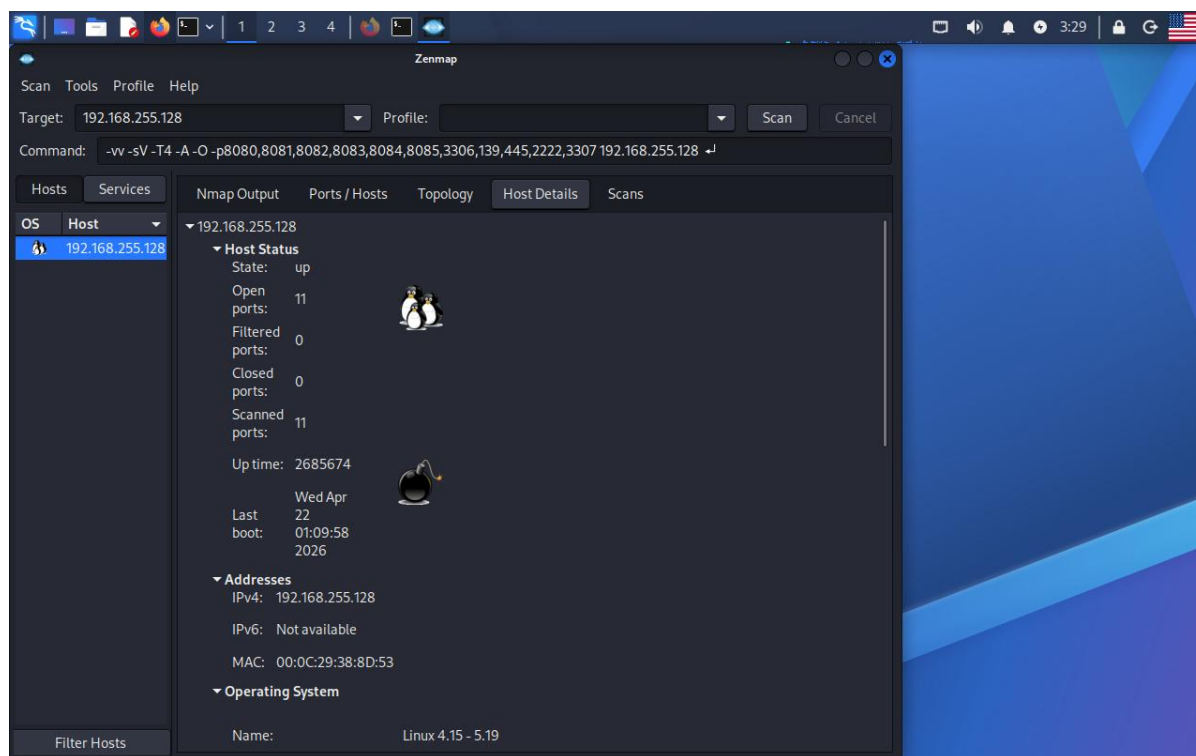


Figure 19 — Zenmap Host Details — Host Status section: State: up, 11 open ports, 0 filtered, 0 closed, 11 scanned. Addresses: IPv4 192.168.255.128, MAC 00:0C:29:38:8D:53 (VMware). Uptime: 2,685,674 seconds (~31 days). Last boot: Wed Apr 22 01:09:58 2026. OS: Linux 4.15–5.19.

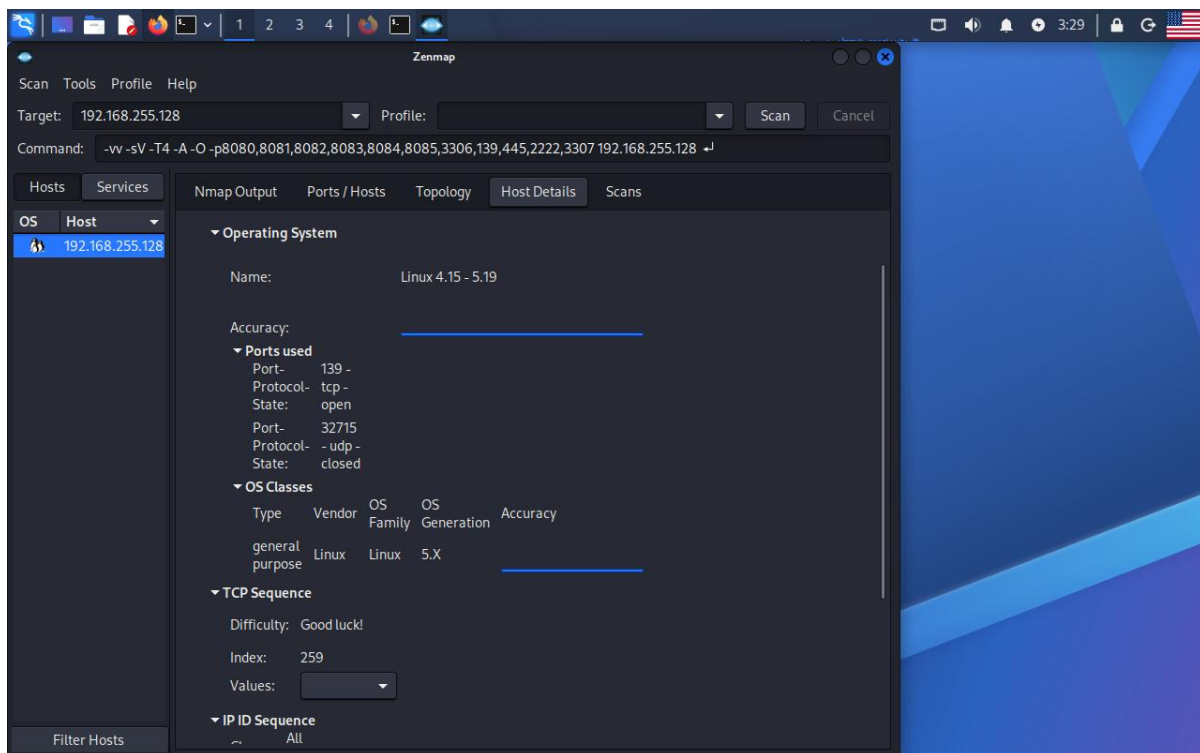


Figure 20 — Zenmap Host Details — Operating System section: OS Name Linux 4.15–5.19, OS Family Linux, Generation 5.x. TCP Sequence Difficulty: "Good luck!" (Index 259, indicating highly random ISNs). IP ID Sequence: All zeros.

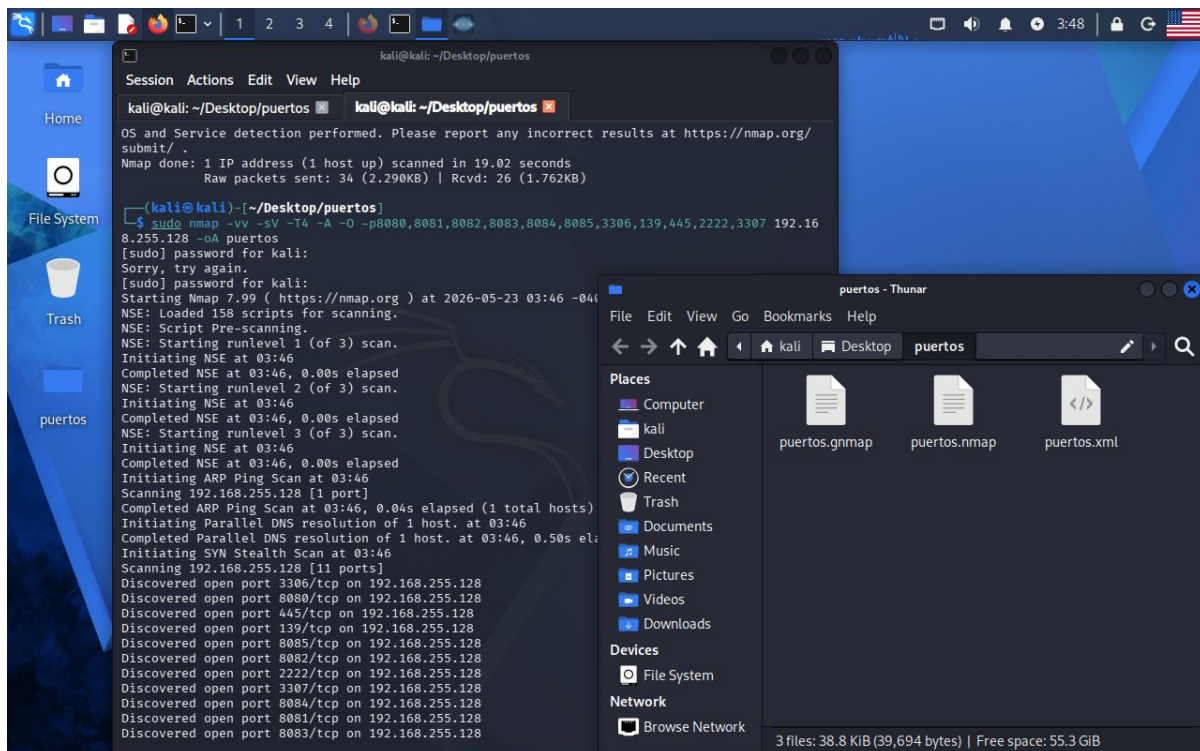


Figure 21 — Zenmap Host Details — continued: TCP TS Sequence Class: 1000HZ. Comments section (empty). This completes the host profiling data collected by the aggressive Nmap scan.

5. Saving and Exporting Scan Results

5.1 — Output Files and HTML Report (puertos)

The scan results were saved using the `-oA puertos` flag, which outputs all three formats simultaneously: `puertos.nmap` (human-readable), `puertos.xml` (machine-readable), and `puertos.gnmap` (grepable). The XML file was then converted to a formatted HTML report using `xsltproc` and opened in Firefox.

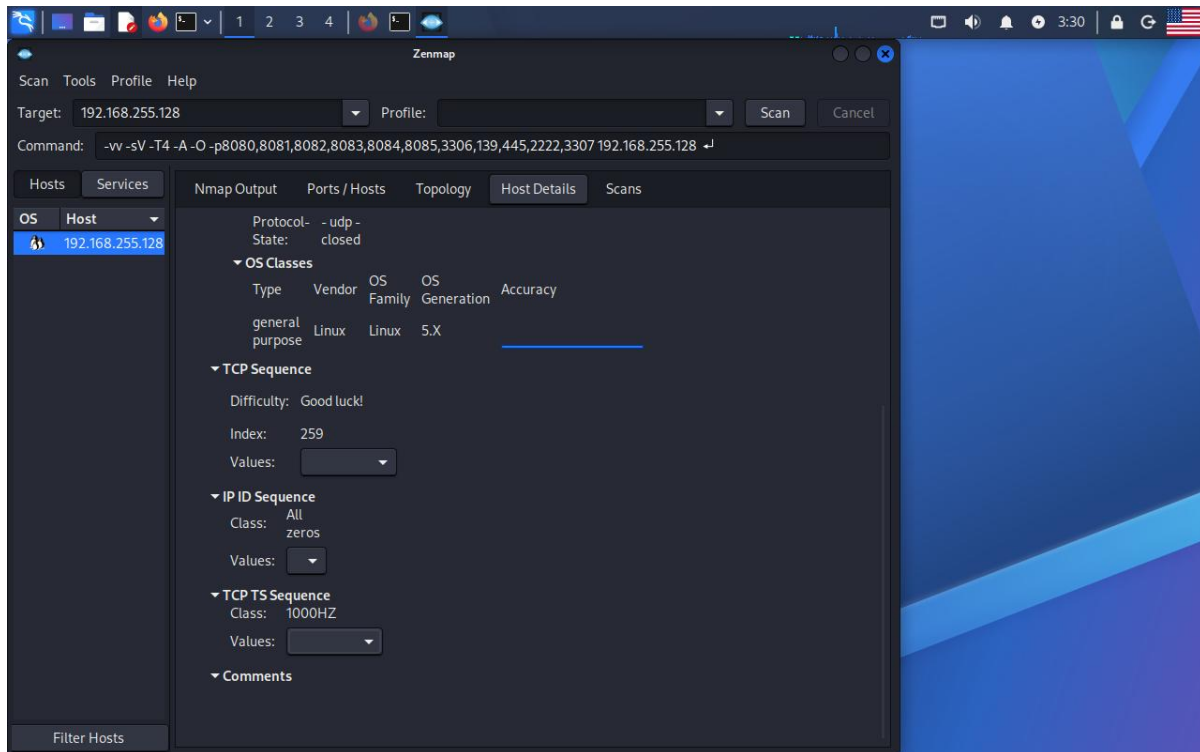


Figure 22 — Kali terminal: second verbose scan run with `-oA puertos` saving all output formats. Thunar file manager shows the `puertos` directory now containing `puertos.gnmap`, `puertos.nmap`, and `puertos.xml` (38.8 KiB total).

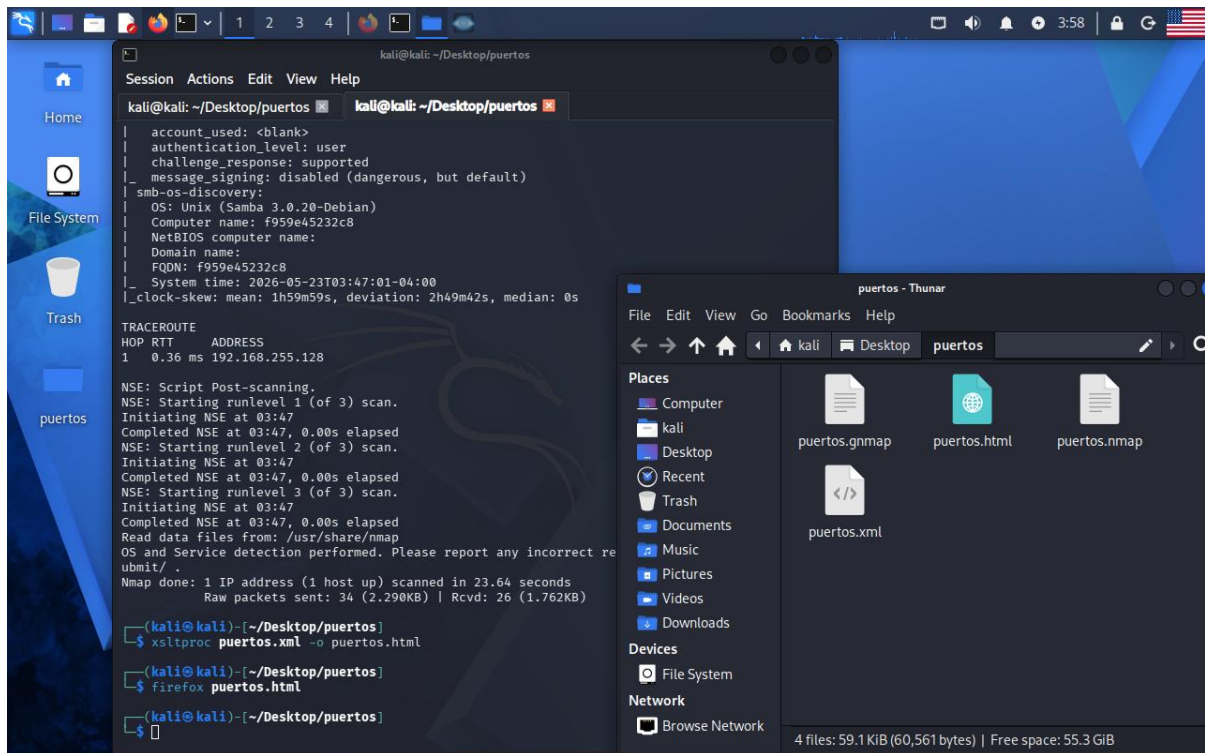
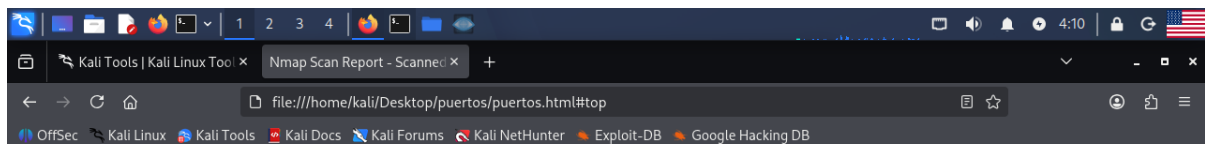


Figure 23 — Kali terminal: `xsltproc puertos.xml -o puertos.html` command converts the XML scan data to an HTML report. Thunar now shows 4 files (puertos.html added). `firefox puertos.html` opens the report in the browser.



Nmap Scan Report - Scanned at Sat May 23 03:46:48 2026

Scan Summary | 192.168.255.128

Scan Summary

Nmap 7.99 was initiated at Sat May 23 03:46:48 2026 with these arguments:
/usr/lib/nmap/nmap -v -sV -T4 -A -O -p8080,8081,8082,8083,8084,8085,3306,139,445,2222,3307 -oA puertos 192.168.255.128
Verbosity: 2; Debug level 0
Nmap done at Sat May 23 03:47:11 2026; 1 IP address (1 host up) scanned in 23.64 seconds

192.168.255.128

Address

- 192.168.255.128 (ipv4)
- 00:0C:29:38:8D:53 - VMware (mac)

Ports

Port	State (toggle closed [0] filtered [0])	Service
139	tcp open	netbios-ssn
445	tcp open	netbios-ssn
2222	tcp open	ssh
ssh-key	1024 60:0f:cf:e1:c0:5f:6a:74:d6:98:24:fa:c4:d5:6:c:d (DSA) ssh-dss AAAAB3NzaC1kc3MAAACBALz4hsc8a2sqn1W960qV8vB0Jc+J17Wxm5METIJH4tKr/xUTwsTYEYnaZLzc0iy21D3Zv0wYb6AA3765zdpCd2Tgand7F0YD5U1XG7b7fz99chReivL05IWEg/E96A1+pqYMP2WD5Ka0 2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA) ssh-rsa AAAAB3NzaC1yc2EAAAABIwAAQEAstqnuFMB0Zv03WTEjP4TudjgWkIVNdTq6kboEDjte0fc65T17sRv0BwqAh0jeeyIk8T55gHdK0D0ak5LSXvLdcmcdYfxeIF0Z5u+nRhiJ7X55A/0c505k3sJ/5Infb78e3anbR	
3306	tcp open	mysql
ssl-date	TLS randomness does not represent time	
ssl-cert	Subject: commonName=MySQL_Server_5.7.44_Auto_Generated_Server_Certificate Issuer: commonName=MySQL_Server_5.7.44_Auto_Generated_CA_Certificate Public Key type: rsa Public Key bits: 2048	

Go to top
Toggle Closed Ports
Toggle Filtered Ports

Figure 24 — Firefox displaying the Nmap Scan Report HTML file (puertos.html): scan summary header "Scanned at Sat May 23 03:46:48 2026", target 192.168.255.128, arguments used, verbosity level 6, and the port table listing 139, 445, 2222, 3306 with their service names and version details.

5.2 — Second Scan Output (sbyv)

A second scan run was performed and saved with a different output filename (`-oA sbyv`) to preserve a separate set of results. This scan was identical in scope and confirmed the same 11 open ports.

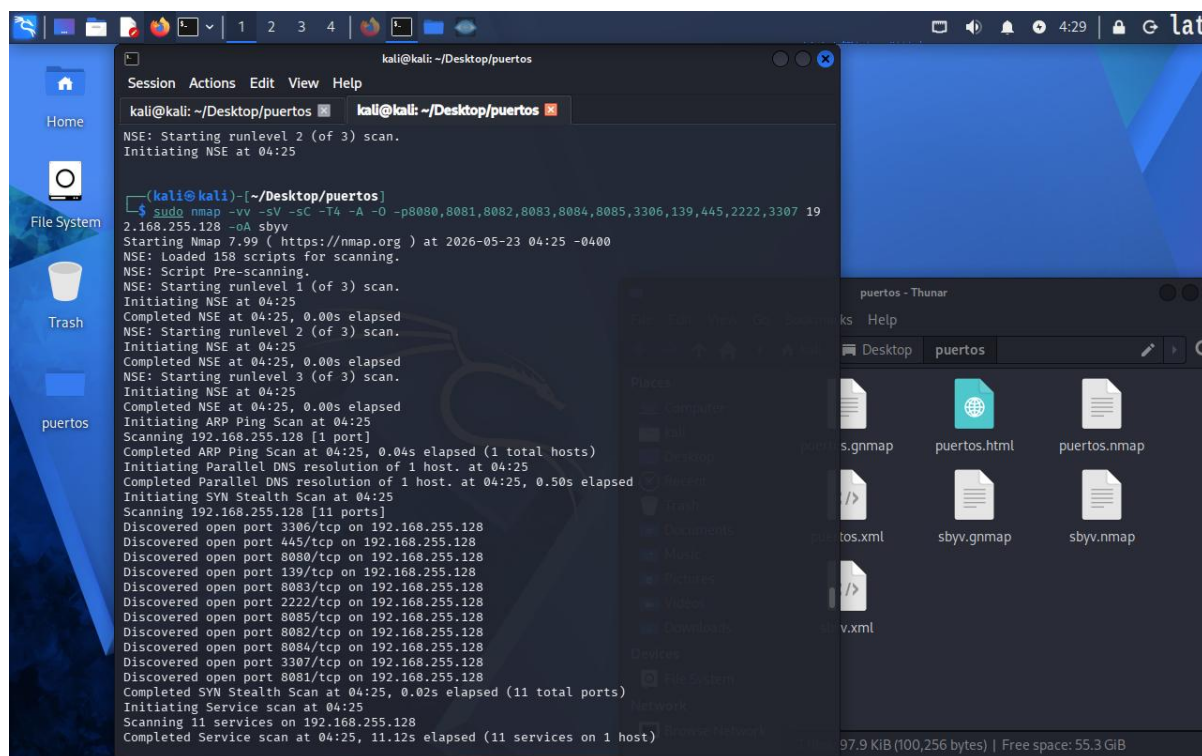


Figure 25 — Kali terminal: third scan run with `-oA sbyv` output. Thunar file manager shows the addition of sbyv.gnmap and sbyv.nmap to the puertos directory (total 7 files, 107.9 KiB).

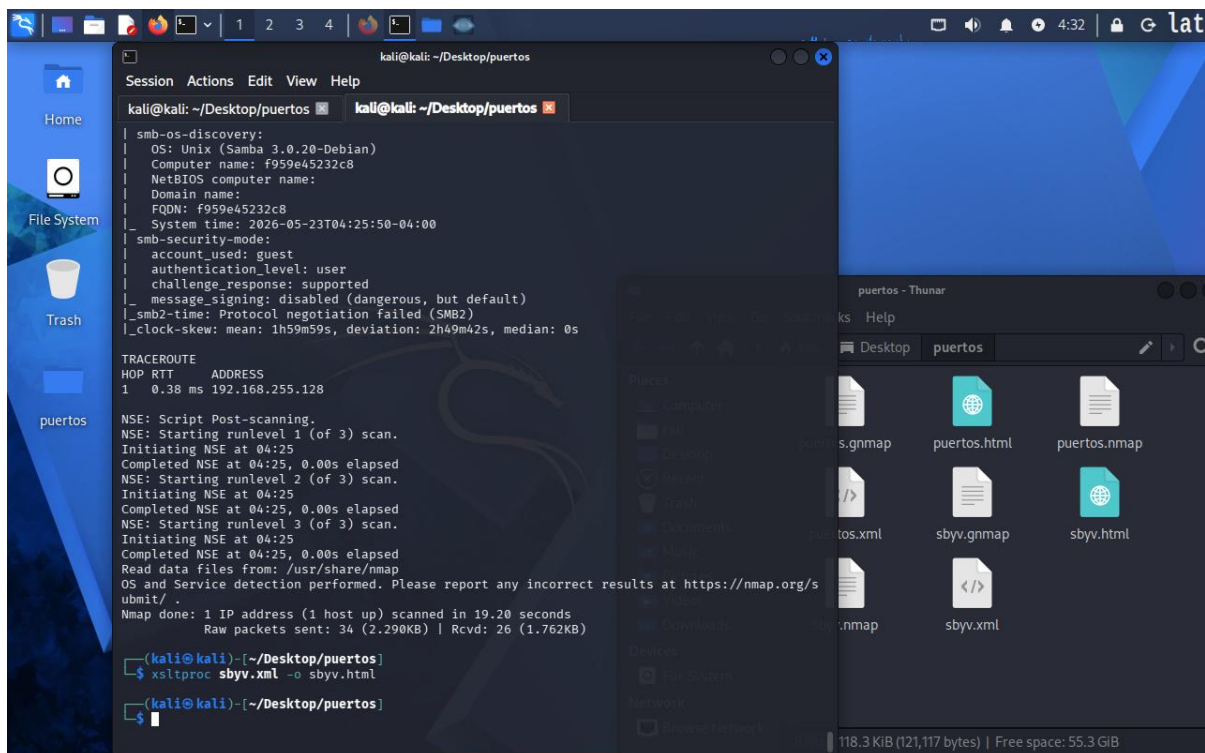
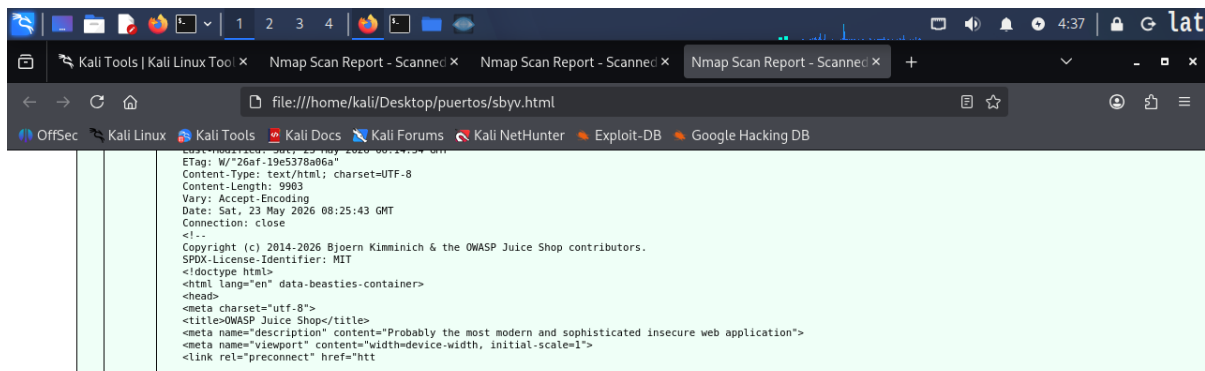


Figure 26 — Kali terminal: scan completes with SMB discovery, traceroute, and NSE post-scanning logs. `xsltproc sbyv.xml -o sbyv.html` generates the second HTML report. Thunar confirms sbyv.html and sbyv.xml are saved (8 files total, 118.3 KiB).



Remote Operating System Detection

- Used port: 139/tcp (open)
- Used port: 41946/udp (closed)
- OS match: Linux 4.15 - 5.19 (100%)
- OS identified but the fingerprint was requested at scan time. (click to expand)

Operating System fingerprint

```
OS: SCAN(V=7, 99%E=4%D=5/23%OT=139%CT=4%CU=41946%PV=Y%D5=1%D=C=D%G=N%M=00C29%T
OS: M=6A116493P=x86 64-pc-linux-gnu)SEQ(SP=106%GCD=1%ISR=108%TI=Z%CI=Z%TS=A
OS: OPS(O1=M5B4ST11NWA02=M5B4ST11NWA03=M5B4NNT11NWA04=M5B4ST11NWA05=M5B
OS: 4ST11NWA06=M5B4ST11)WIN(WL=FEB8W2=FEB8W3=FEB8W4=FEB8W5=FEB8W6=FEB8
OS: )JECN(R=Y%DF=Y%T=3F%W=AF0%Q=M5B4NNSMACC=Y%Q=T1(R=Y%DF=Y%T=3F%Q=0%A=5
OS: %F=AS%RD=0%Q)=T2(R=N)T3(R=N)T4(R=Y%DF=Y%T=3F%W=0%Q=0%A=Z%F=R%Q=0%Q=0)
OS: TS(R=Y%DF=Y%T=40%W=0%Q=Z%A=5%F=AR%Q=0%RD=0%Q)=T6(R=Y%DF=Y%T=40%W=0%Q=0%A
OS: =Z%F=R%Q=0%RD=0%Q)=T7(R=Y%DF=Y%T=40%W=0%Q=Z%A=5%F=AR%Q=0%RD=0%Q)=U1(R=Y%Q
OS: F=H%T=40%IPL=164%UN=0%RIPL=G%RID=G%RIPCK=G%RUCK=G%RUD=G)G(R=Y%DF=Y%T=4
OS: 0%Q=0)CD=5)
```

Host Script Output

Script Name	Output
smb2-security-mode	Couldn't establish a SMBv2 connection.
inOn.mnlinker	

Go to top
Toggle Closed Ports
Toggle Filtered Ports

Figure 27 — Firefox displaying the second Nmap Scan Report (sbyv.html): Remote Operating System Detection section confirming Linux 4.15–5.19 (100% accuracy), OS fingerprint string, and Host Script Output table showing smb2-security-mode result: "Couldn't establish an SMBv2 connection."

6. Summary of Findings

The Nmap reconnaissance phase successfully profiled the Metasploitable2 target and produced the following findings:

- Target IP: 192.168.255.128 (MAC: 00:0C:29:38:8D:53, VMware adapter)
- Operating System: Linux 4.15 – 5.19 (Linux kernel, general purpose, OS fingerprint confidence ~100%)
- Last boot: April 22, 2026 | Uptime: ~31 days
- 11 open TCP ports discovered: 139, 445, 2222, 3306, 3307, 8080, 8081, 8082, 8083, 8084, 8085
- SMB: Samba 3.0.20-Debian — message signing disabled (dangerous default), SMBv2 not available
- SSH: OpenSSH 4.7p1 (very old, Debian, protocol 2.0) on port 2222
- MySQL: 5.7.44 (port 3306) and 5.0.51a-3ubuntu5 (port 3307 — the Metasploitable2 version)
- HTTP services: nginx 1.31.0, DVWA (Apache 2.4.25), Nagios NSCA, WordPress (Apache 2.4.67), Apache DAV/2, OWASP Juice Shop
- TCP Sequence Difficulty: "Good luck!" — strong ISN randomization
- Scan results exported in .nmap, .xml, .gnmap, and .html formats

These findings provide a complete attack surface map for the next phases of the lab, where tools such as Metasploit, Hydra, SQLmap, or Burp Suite will be used to actively exploit the identified vulnerabilities.